

Osman Melih Cabik

Saarbrücken, Germany

✉ osmanmelih.cabik@gmail.com | 🌐 osmmelih.github.io | 🌐 osmanmelihcabik | 🌐 osmmelih

ABOUT ME

M.Sc. student in Educational Technology at Saarland University specializing in Human-Computer Interaction (HCI), C++/C# software development, and interactive 3D/VR/AR simulation UI systems. Experienced in engineering real-time AR/GPS HUD interfaces, simulation game mechanics in Unreal Engine 5 & Unity 3D (Meta XR SDK), and multi-platform software tools. Passionate about contributing XR software infrastructure, spatial interaction mechanics, and clean code practices to cutting-edge research projects.

EDUCATION

Universität des Saarlandes

Saarbrücken, Germany

M.Sc. Educational Technology — Current GPA: 1.7 / 1.0

Oct 2025 – Present

- **Core Focus:** Human-Computer Interaction (HCI), Interactive Systems, Spatial Learning Analytics, Software UI Accessibility.

Istanbul University-Cerrahpasa

Istanbul, Türkiye

Bachelor of Education in Preschool Teaching — Final Grade: 3.64 / 4.0 (~1.36 German Grade)

Sep 2018 – Jun 2022

- **Core Focus:** Instructional Design, Cognitive Load Theory, Human-Centered Design.

EXPERIENCE & ACADEMIC PROJECTS

Land of Chasers

Ankara, Türkiye

Founder & Simulation UI / C++ & Unity Developer

Oct 2022 – Jan 2026

- Engineered interactive 3D simulation mechanics and microlearning games using C++, **Unreal Engine 5**, and **Unity 3D**, optimizing real-time rendering performance and UI responsiveness.
- Designed cross-platform digital asset pipelines and user control interfaces, applying HCI principles to minimize cognitive load during real-time 3D interaction.
- Applied modular design patterns and version control (Git) to maintain scalable codebase architecture and user telemetry tracking.

Antigravity 2.0: GPS-Powered AR Scavenger Hunt

Saarbrücken, Germany

AR Software & UI Developer (Dr. Kristin Altmeyer's Course Project)

2025

- Developed a full-stack, GPS-powered outdoor AR learning platform utilizing **React 18**, **Leaflet**, and **HTML5 MediaDevices API** for real-world coordinate tracking.
- Engineered a live AR camera overlay with diegetic HUD interfaces, context-adaptive tasks, and real-time student leaderboard mechanics.

AI-Supported Interactive 3D VR Spatial Representation System

Saarbrücken, Germany

VR / XR Systems Developer

2025

- Designed an interactive 3D spatial geometry & interaction framework using **Unity 3D**, **Meta XR SDK (OpenXR)**, and **SymPy** for real-time spatial manipulation.

Google - AI and Tech Academy

Ankara, Türkiye

Product Owner Trainee & Cross-Platform Developer

Nov 2022 – Jul 2023

- Co-developed EduCon EdTech mobile platform (**Flutter**, **Firebase**), leading Agile/Scrum sprints, backlog prioritization, and UI/UX usability reviews.

ESSA (Virtual Home Classroom Platform)

Istanbul, Türkiye

Instructional Designer & E-Learning Developer

Jan 2021 – Sep 2021

- Developed interactive digital learning interfaces serving 1,000+ active users; recognized as EdTech Finalist at **TEKNOFEST 2021**.

SKILLS & LANGUAGES

Programming & UI Frameworks: C#, C++, Python, TypeScript, Flutter, HTML/CSS, Qt/QML (Basics).

VR, AR & Simulation Tools: Unity 3D (Meta XR SDK, OpenXR, Hand Tracking), Unreal Engine 5, Antigravity 2.0 AR, MediaDevices API, Leaflet Maps, Figma, Articulate Storyline.

Infrastructure & DevOps: Git, CMake / Build Toolchains, Firebase, Supabase (PostgreSQL), Agile & Scrum, Jira, Confluence.

Languages: English (C1 - Fluent), German (B1 - Intermediate), Turkish (Native).